AI Capstone Final Project Report - Car Accident Detection

Team ID: E4
111550072林祐晨　　111550116王紹安

1 Motivation

Traffic scene understanding is an important part of computer vision that focuses on interpreting dynamic road environments. It encompasses tasks such as object detection (e.g., vehicles, pedestrians), lane detection, traffic sign recognition, and behavior prediction. And we chose accident detection as topic for several key reasons:

1. **High social impact**: Traffic accidents are serious incidents with potentially severe consequences. Accident detection can help accelerate emergency responses and reduce casualties, offering significant value for public safety.
2. **Clear real-world applications**: This technology can be directly applied to smart city infrastructure, road surveillance systems, dashcam analytics, and autonomous vehicle incident response, making it highly practical and relevant.
3. **Challenging**: Accident detection involves object detection and scene understanding. It requires integrating multiple computer vision techniques, making it an ideal project to deepen our understanding of deep learning models like YOLO.
4. **Accessible data and clear annotations**: Several publicly available datasets provide labeled accident footage, which makes training and validating the model more feasible and efficient.

2 Techniques Used

2.1 YOLO [1][2]

2.1.1 Overview

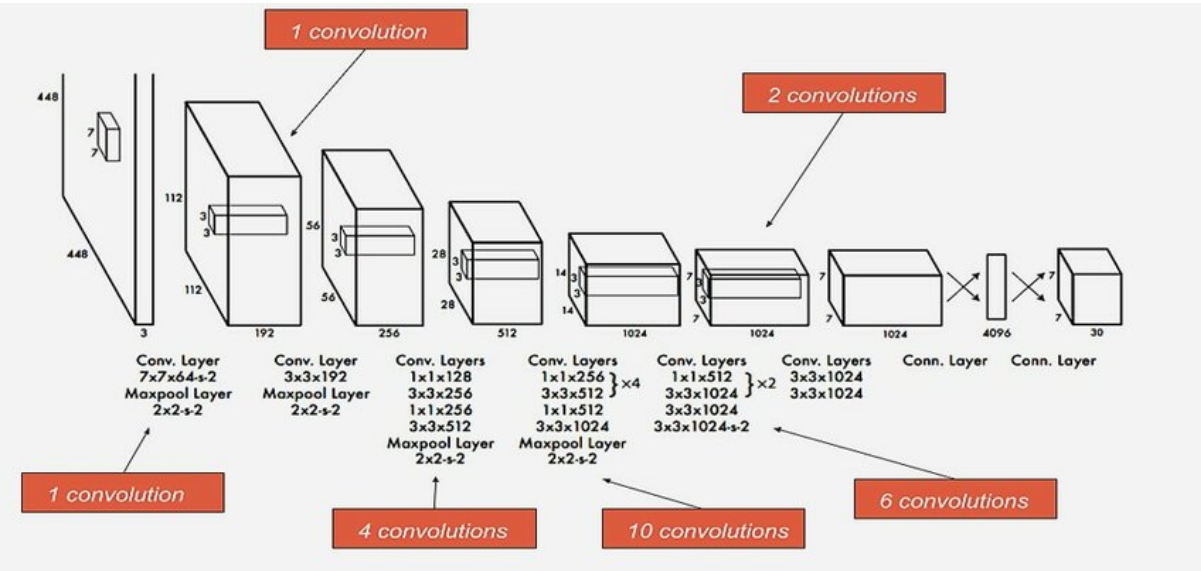
YOLO unifies the traditionally separate steps of object detection into a single neural network. It predicts all bounding boxes across an image using the extracted features of the entire image, and simultaneously estimates the probability of each class for every bounding box. Each input image is divided into an SxS grid. If the center of an object falls into a particular grid cell, that cell is responsible for detecting the object. The confidence score reflects two things: the confidence that a bounding box contains an object, and the accuracy of the bounding box in localizing that object.

$$conf.\ score = Pr(Object) * IOU\ (groundtruth)$$

Each bounding box has five predicted parameters: **x, y, w, h, and confidence**. The coordin ates (x, y) represent the offset of the bounding box center relative to the grid cell. The width (w) an d height (h) indicate the size of the bounding box. The **confidence** score represents the Intersection over Union (IoU) between the predicted box and any ground truth box. Additionally, each grid cell predicts the **conditional class probabilities** for all possible classes, indicating the likelihood of eac h class being present in that cell. During inference, these class probabilities are multiplied by the c onfidence score of each bounding box to obtain the final class-specific confidence scores.

$$Pr(Class\text{-}i|Object) * Pr(Object) * IOU = Pr(Class\text{-}i) * IOU$$

2.1.2 Network Design



This diagram illustrates the full architecture of the YOLO. The core idea is to pass the enti re input image through a single convolutional neural network that predicts all bounding boxes and class probabilities. The input image goes through several convolutional and max pooling layers to extract visual features. The early stage includes one large convolution followed by multiple small c onvolutions and pooling operations, gradually transforming spatial details into high-level features.

After this deep feature extraction, the resulting feature map is passed through two fully con nected layers. It departs from traditional object detection pipelines that separate region proposal an d classification stages, offering an integrated and efficient inference strategy.

2.1.3 Training Metric

In CNN, the final layer uses a linear activation function, while the other neurons use the Le aky ReLU activation function:

$$\phi(x) = \begin{cases} x, & \text{if } x > 0 \\ 0.1x, & \text{otherwise} \end{cases}$$

The loss function used is the sum-squared error, as it is relatively easier to optimize. Howe ver, during the optimization process, it weights localization errors equally, which may prevent the model from achieving the best possible results. Additionally, during training, many grid cells do no t contain any objects, causing their confidence scores to approach zero. This leads to the errors fro m grid cells that do contain objects being amplified, which can make the model unstable and poten tially cause it to diverge during training.

Finally, during training, the main objective is to optimize the following loss function:

$$
\begin{aligned}
&\lambda_{\textbf{coord}} \sum_{i=0}^{S^2} \sum_{j=0}^{B} \mathbb{1}_{ij}^{\text{obj}} \left[(x_i - \hat{x}_i)^2 + (y_i - \hat{y}_i)^2 \right] \\
&+ \lambda_{\textbf{coord}} \sum_{i=0}^{S^2} \sum_{j=0}^{B} \mathbb{1}_{ij}^{\text{obj}} \left[\left(\sqrt{w_i} - \sqrt{\hat{w}_i} \right)^2 + \left(\sqrt{h_i} - \sqrt{\hat{h}_i} \right)^2 \right] \\
&+ \sum_{i=0}^{S^2} \sum_{j=0}^{B} \mathbb{1}_{ij}^{\text{obj}} \left(C_i - \hat{C}_i \right)^2 \\
&+ \lambda_{\text{noobj}} \sum_{i=0}^{S^2} \sum_{j=0}^{B} \mathbb{1}_{ij}^{\text{noobj}} \left(C_i - \hat{C}_i \right)^2 \\
&+ \sum_{i=0}^{S^2} \mathbb{1}_{i}^{\text{obj}} \sum_{c \in \text{classes}} (p_i(c) - \hat{p}_i(c))^2
\end{aligned}
$$

3 Dataset

3.1 Roboflow [3]

This dataset is obtained from Roboflow, a widely adopted platform for managing and distri buting datasets for computer vision applications. Roboflow hosts a vast collection of publicly avail able datasets contributed by users across the globe. It serves as a centralized hub where researchers and developers can access high-quality, annotated image data to accelerate the development of mac hine learning models.

3.2 Car Accident Dataset

This dataset consists of over 3,000 high-resolution images depicting a wide range of traffic accident scenarios. These images capture real-world incidents involving various types of vehicles s uch as passenger cars, trucks, motorcycles, and buses, under diverse environmental and road condit

ions. The scenes vary in complexity, including single-vehicle accidents, multi-vehicle collisions, an d instances involving roadside infrastructure or pedestrians.

The dataset reflects significant variation in lighting and weather conditions, encompassing i mages taken during the day and night, as well as in rain, fog, and other adverse weather scenarios. This diversity contributes to the robustness of the dataset, making it well-suited for training models that need to perform reliably in real-world applications.

3.3 Labeling and Annotation Method

Each image in the dataset has been systematically annotated using bounding boxes that pre cisely delineate the regions affected by traffic accidents. These annotations are not limited to appro ximate visual cues; rather, they are defined by the exact coordinates of the four corner points of ea ch bounding box. This allows for pixel-level spatial accuracy in identifying the location and extent of the accident area within each frame.

Moreover, the annotations are formatted in a way that ensures compatibility with a variety of popular machine learning frameworks and architectures, such as YOLO in our case.

4 Experiment

4.1 Data Partition

The dataset was systematically divided into three subsets: a training set containing 2,415 images, a validation set with 338 images and a test set comprising 339 images. Such a well-structured data split minimizes the risk of overfitting and ensures a more reliable and objective assessment of model performance, which is particularly crucial in high-stakes applications like accident detection and automated traffic analysis.

4.2 Training

In this project, YOLOv11n—an efficient, lightweight variant of YOLO was utilized as the backbone for detecting traffic accidents. The model was initialized with pretrained weights, enabling transfer learning from prior knowledge(e.g. car detection) and accelerating convergence during training.

Each input image is resized to 640×640 pixels prior to being fed into the model. The YOLO family of models is designed to operate on square image inputs that are multiples of 32, due to the downsampling nature of the architecture. An image size of 640 satisfies this constraint (640 ÷ 32 = 20), ensuring that feature maps align correctly throughout the convolutional layers. The processing plays a crucial role in determining both the detection performance and the computational efficiency of the model.

4.3 Result

The following is the overview of the result prediction compared to the ground truth of the dataset.

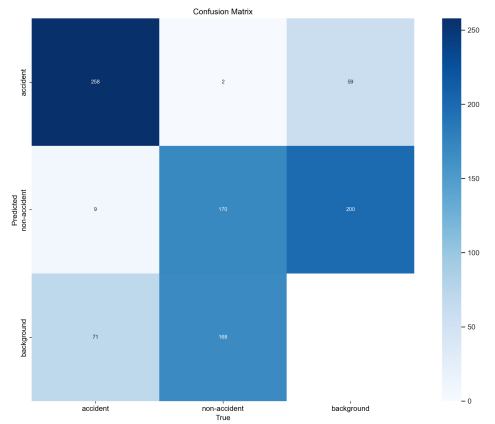
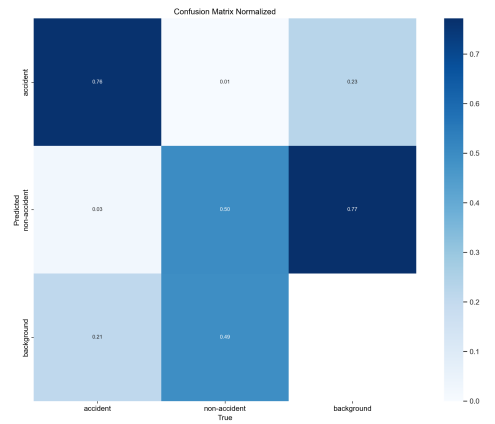
Ground Truth	Prediction

Overall, the model demonstrated high reliability in identifying accident instances, with the predicted bounding boxes closely aligning with the annotated ground truth regions in the majority of cases. While there were minor discrepancies—such as slight shifts in bounding box coordinates or variations in box dimensions—these differences were generally small and did not compromise t he correctness of the detection outcome.

4.4 Matrices Evaluation

4.4.1 Confusion Matrix

The confusion matrix shows the classification performance across three classes: accident, n on-accident, and background. Each row corresponds to the predicted class, and each column corres ponds to the true class.

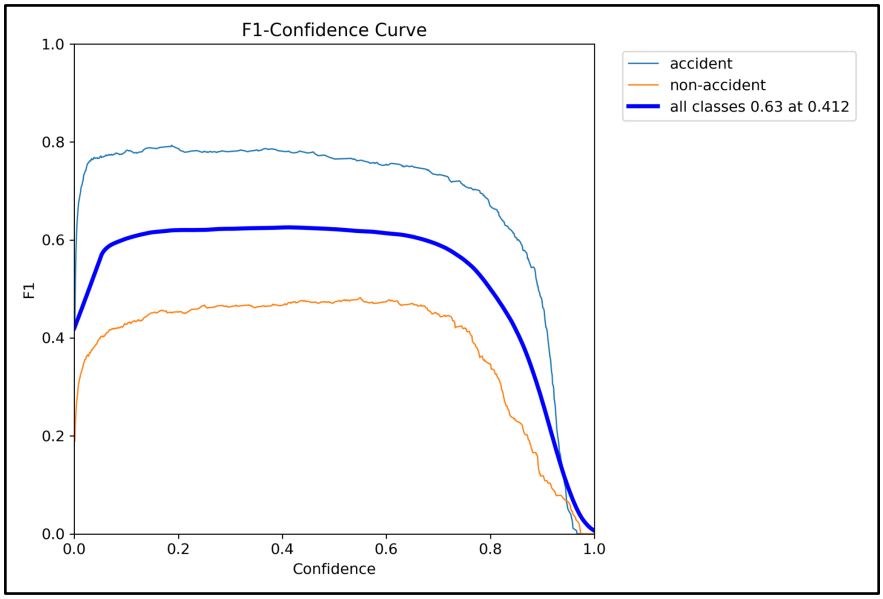
Confusion Matrix	Normalized Confusion Matrix
	

The model performs well on the accident and background classes, especially with high pre cision for "background" (77%) and accident (76%). Non-accident is the most confusing class, with only 50% correctly classified and a large portion (47%) being mistaken for background. There is a noticeable confusion between non-accident and background, which suggests these two classes may have similar features or insufficient separation in the model, as we can see in section 4.3.

4.4.2 F1 Curve

This plot shows the F1 score as a function of prediction confidence thresholds for each clas s:accident (blue), non-accident (orange) and all classes (macro average) — thick blue line.

Best overall F1 score is 0.63, achieved at a confidence threshold of 0.412. This is a good re ference point for setting a classification threshold. Accident class achieves the highest F1 score, co nsistently above 0.75 across a wide confidence range (~0.1 to ~0.7), indicating the model is quite g ood at distinguishing accidents. Non-accident class shows the weakest performance, with an F1 sc ore around 0.45~0.5 for most confidence levels. It suggests the model struggles with precision or r ecall for this class. All curves drop sharply when confidence exceeds ~0.8, meaning very confident predictions tend to be incorrect — likely due to overconfidence in a small number of misclassified samples.



5 Learned from project

Through this project, we gained a deeper and more comprehensive understanding of the Y OLO model. We learned about its core design philosophy, which integrates the traditional object d etection pipeline into a single neural network that allows for end-to-end training and inference. We understood how YOLO divides the input image into grids, with each grid cell predicting multiple b ounding boxes and class probabilities, and how the confidence score reflects both the presence of a

n object and the accuracy of the prediction. By analyzing the network architecture, we learned how each convolutional layer contributes to the model, and how different activation functions such as leaky ReLU and linear activation are applied. We also studied the sum-squared error used as the loss function in YOLO, including its benefits and how it might cause instability during training—especially when many grid cells contain no objects.

Finally, by implementing a traffic accident detection task on video data using YOLO, We practiced model training, exporting, inference, and visualization. This experience not only enhanced my theoretical knowledge but also strengthened my hands-on skills in real-world applications.

6 Future Work

In the future, we plan to further improve our traffic accident detection system in several ways. First, we aim to fine-tune the model using a larger and more diverse dataset that includes a broader range of accident scenarios, lighting conditions, and traffic environments. This can help the model generalize better to real-world applications. Second, we intend to explore the use of more advanced YOLO variants, such as YOLOv8 or custom lightweight architectures for deployment on edge devices like surveillance cameras. Third, we hope to integrate temporal information by using models like ConvLSTM or 3D CNNs to capture motion patterns over time, potentially improving detection accuracy in complex scenes. Lastly, incorporating a real-time alert system based on prediction results would be a practical extension, allowing quick response in smart traffic management systems.

7 Contributions

林祐晨: Model training, report
王紹安: Model training, report

Reference

[1]https://medium.com/@c.c.lo/yolo-%E4%BB%8B%E7%B4%B9-4307e79524fe
[2]https://www.researchgate.net/figure/YOLO-Full-Model-Architecture_fig4_371136193
[3]https://universe.roboflow.com/accident-and-nonaccident/accident-and-non-accident-label-image-dataset